



Question	Answer	Marks	Guidance
1(a)		B1	Symmetrical. In correct position. Lines intended to be straight. Must be in both first and second quadrants. Key coordinates must be correct. Ignore $y = x + 5a$ if seen.
		1	

Question	Answer	Marks	Guidance
1(b)	Obtain critical value $\frac{7a}{2}$ from $x + 5a = 3x - 2a$	B1	Allow if seen in an inequality.
	Obtain critical value $-\frac{3a}{4}$ from $x + 5a = 2a - 3x$	B1	Allow if seen in an inequality.
	State final answer $-\frac{3a}{4} < x < \frac{7a}{2}$	B1	SC B1 only for $-\frac{3a}{4} < x < \frac{7a}{2}$ with <i>their a</i> from part (a). Allow any equivalent notation. Allow $-\frac{3a}{4} < x$ and $x < \frac{7a}{2}$.
	Alternative Method for Question 1(b)		
	Solve quadratic equation $(3x - 2a)^2 = (x + 5a)^2$	M1	$8x^2 - 22ax - 21a^2 = 0$
	Obtain critical values $-\frac{3a}{4}$ and $\frac{7a}{2}$	A1	
	State final answer $-\frac{3a}{4} < x < \frac{7a}{2}$	A1	SC B1 only for $-\frac{3a}{4} < x < \frac{7a}{2}$ with <i>their a</i> from part (a). Allow any equivalent notation. Allow $-\frac{3a}{4} < x$ and $x < \frac{7a}{2}$.
		3	

Question	Answer	Marks	Guidance
2	Use the correct rule for logarithm of a power, product or quotient or an equivalent method using exponentials	*M1	Use of any correct law applied to original terms.
	Obtain an equation free of logarithms	A1	E.g. $\frac{(2x+3)^2}{2x+5} = 3x$.
	Form a 3-term quadratic from completely correct use of logarithms and solve for x	DM1	$2x^2 + 3x - 9 = 0$
	State final answer $\frac{3}{2}$ only	A1	OE Rejection of negative value if given must be clear.
		4	

Question	Answer	Marks	Guidance
3(a)	Express the left hand side in terms of $\sin \theta$ and $\cos \theta$	M1	$\frac{\cos^2 \theta}{\sin^2 \theta} - \frac{\sin^2 \theta}{\cos^2 \theta}$
	Combine to a single term and factorise the numerator	M1	$\frac{(\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta)}{\cos^2 \theta \sin^2 \theta}$
	Use correct double angle formulae in numerator and denominator	M1	$\frac{4 \cos 2\theta}{\sin^2 2\theta}$
	Obtain $4 \cot 2\theta \operatorname{cosec} 2\theta$ from correct working	A1	$\frac{4 \cos 2\theta}{\sin 2\theta} \times \frac{1}{\sin 2\theta}$ or $\frac{4 \cos 2\theta}{\sin 2\theta \sin 2\theta}$.
	Alternative Method for Question 3(a)		
	Express the right hand side in terms of $\sin 2\theta$ and $\cos 2\theta$ or $\tan 2\theta$ and $\sin 2\theta$	M1	$\frac{4 \cos 2\theta}{\sin 2\theta} \times \frac{1}{\sin 2\theta}$ or $\frac{4}{\tan 2\theta} \times \frac{1}{\sin 2\theta}$ or $\frac{4 \cos 2\theta}{\sin 2\theta \sin 2\theta}$.
	Use correct double angle formulae in numerator and denominator	M1	$\frac{(\cos^2 \theta - \sin^2 \theta)}{\sin^2 \theta \cos^2 \theta}$ or $\frac{4(1 - \tan^2 \theta)}{2 \tan \theta} \times \frac{1}{2 \sin \theta \cos \theta}$ OE
	Split into 2 terms and simplify	M1	$\operatorname{cosec}^2 \theta - \sec^2 \theta$
	Obtain $\cot^2 \theta - \tan^2 \theta$ from correct working	A1	$(1 + \cot^2 \theta) - (1 + \tan^2 \theta)$

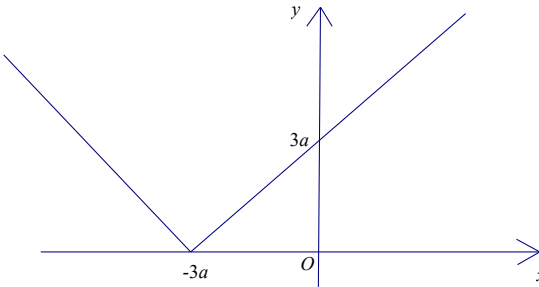
Question	Answer	Marks	Guidance
3(a)	Alternative Method 2 for Question 3(a)		
	Express the left hand side as a difference of 2 squares in terms of $\tan \theta$	M1	$\left(\frac{1}{\tan \theta} - \tan \theta\right) \times \left(\frac{1}{\tan \theta} + \tan \theta\right)$
	Use correct formula for $\tan 2\theta$	M1	$\left(\frac{2}{\tan 2\theta}\right) \times \left(\frac{1}{\tan \theta} + \tan \theta\right)$
	Use $1 + \tan^2 \theta = \sec^2 \theta$ and simplify using correct double angle formula	M1	$\left(\frac{2}{\tan 2\theta}\right) \times \left(\frac{\sec^2 \theta}{\tan \theta}\right)$
	Obtain $4 \cot 2\theta \operatorname{cosec} 2\theta$ from correct working	A1	$\left(\frac{2}{\tan 2\theta}\right) \times \left(\frac{2}{\sin 2\theta}\right)$
	Alternative Method 3 for Question 3(a)		
	Express the left hand side using appropriate identities	M1	$\cot^2 \theta = \operatorname{cosec}^2 \theta - 1$ and $\tan^2 \theta = \sec^2 \theta - 1$ leading to $\operatorname{cosec}^2 \theta - \sec^2 \theta$.
	Combine to a single term in terms of $\sin \theta$ and $\cos \theta$	M1	$\frac{(\cos^2 \theta - \sin^2 \theta)}{\sin^2 \theta \cos^2 \theta}$
	Use correct double angle formulae in numerator and denominator	M1	$\frac{4 \cos 2\theta}{\sin^2 2\theta}$
	Obtain $4 \cot 2\theta \operatorname{cosec} 2\theta$ from correct working	A1	$\frac{4 \cos 2\theta}{\sin 2\theta} \times \frac{1}{\sin 2\theta}$ or $\frac{4 \cos 2\theta}{\sin 2\theta \sin 2\theta}$.
		4	

Question	Answer	Marks	Guidance
3(b)	Use the identity to obtain an expression in one trigonometric function	*M1	
	Obtain $\tan^2 2x = \frac{4}{5}$	A1	OE
	Obtain one solution e.g. 20.9°	DM1	AWRT
	Obtain a second value e.g. 69.1° and no extras in range	A1	AWRT
		4	

Question	Answer	Marks	Guidance
4	Form a quadratic equation in 2^x	M1*	Need not have all terms on one side. Allow 2^{2x} but not $2^x \times 2^x$.
	Obtain $4 \times 2^2 \times (2^x)^2 - 3 \times 2^x - 5 \times 4 = 0$ Needs $(2^x)^2$ but may be implied if progressing to solve the quadratic	A1	Or three-term equivalent, e.g. $(2^{x+4})^2 - 3 \times 2^{x+4} - 320 = 0$. Terms on LHS can be in any order, e.g. $16 \times (2^x)^2 - 3 \times 2^x - 20 = 0$.
	Solve for x (do not need to solve quadratic correctly for this mark) Allow, e.g., $x \ln 2 = \ln \dots$ to score M1 at this stage	DM1	Requires correct method for solving $2^x = k$ ($2^x = 1.2157 \dots$). Need to see some working, such as $x \ln 2 = \ln 1.2157$ or $\log_2(1.2157)$.
	Obtain $x = 0.282$ only final answer	A1	No working at all scores M0*A0M0A0 . From a quadratic equation directly to correct answer scores M1*A1M0A0 . From solution to quadratic to correct answer M1*A1M0A0 , then SCB1 .
		4	

Question	Answer	Marks	Guidance
5(a)	Complete division by $x^2 + x + 1$ as far as quotient $= 2x^2 + kx + \dots$	M1	Note: The -3 is obtained by comparing the constant terms. Alternative by completing the division gives the quotient as $2x^2 + (a-2)x + (4-a)$.
	$ \begin{array}{r} 2x^2 \quad + (a-2)x \quad -3 \\ x^2 + x + 1 \overline{) 2x^4 \quad \quad ax^3 \quad \quad 4x^2 \quad \quad bx \quad -3} \\ \underline{2x^4 \quad \quad 2x^3 \quad \quad 2x^2} \\ (a-2)x^3 \quad \quad 2x^2 \\ \underline{(a-2)x^3 \quad (a-2)x^2 \quad (a-2)x} \\ (4-a)x^2 \quad (b-a+2)x \quad -3 \\ \underline{-3x^2 \quad \quad -3x \quad -3} \end{array} $		
	Equate (part of the) remainder to 0 and solve for a or b	M1	
	Obtain $a = 7$	A1	E.g. from $4 - a = -3$.
	Obtain $b = 2$	A1	E.g. from $b - a + 2 = 4 - a$.
	Alternative Method for Question 5(a)		
	State $p(x) = (x^2 + x + 1)(2x^2 + px - 3)$	B1	By inspection.
	Compare coefficients to obtain a or b	M1	$2 - 3 + p = 4$, $a = 2 + p$, $b = -3 + p$
	Obtain $a = 7$	A1	
	Obtain $b = 2$	A1	
		4	

Question	Answer	Marks	Guidance
5(b)	Obtain or substitute $x = -3$ into <i>their</i> quotient or into original expression	M1	Allow with a and b non-numerical.
	Must evaluate $(-3)^4$ etc. when substituting, and reach $= 0$ Conclude that $x + 3$ is a factor Statement needed either initially or finally	A1	AG CWO Need not evaluate, e.g. 2×81 .
	Alternative Method for Question 5(b)		
	Factorise correct quotient to obtain $(2x - 1)(x + 3)$	M1	Implied by $(x^2 + x + 1)(2x - 1)(x + 3)$.
	Obtain $(x^2 + x + 1)(2x - 1)(x + 3)$, or ‘therefore, $(x + 3)$ is a factor of $p(x)$ ’	A1	AG CWO
		2	

Question	Answer	Marks	Guidance
6(a)		B1	Roughly symmetrical. Condone some inaccuracy, but needs to look as if they intended symmetry and straight lines. If not, then B0. Needs to be in the correct position. Needs to have two solid line segments. Needs to exist in both quadrants above the axis. Ignore dotted lines below the axis. Solid line below the axis is B0. Condone if no scale shown, but need to see $3a$ and $-3a$ marked. Ignore $y = a - 2x$ if seen.
		1	
6(b)	Obtain critical value $-\frac{2a}{3}$ from $x + 3a = a - 2x$	B1	Ignore $x = 4a$ if seen.
	State final answer $x > -\frac{2a}{3}$	B1	Need a clear conclusion – must imply rejection of $x = 4a$. B0 if using $\geq$.
	Alternative Method for Question 6(b)		
	Obtain critical value $-\frac{2a}{3}$ from $(x + 3a)^2 = (a - 2x)^2$	B1	Ignore $x = 4a$ if seen.
	State final answer $x > -\frac{2a}{3}$	B1	Need a clear conclusion – must imply rejection of $x = 4a$. B0 if using $\geq$.
		2	

Question	Answer	Marks	Guidance
7	Use a correct law for the logarithm of a product or the logarithm of a power	*M1	Not available after incorrect manipulation, e.g. $6^{x+1} = 12^{2x-3}$
	Obtain $\ln 3 + (x+1)\ln 2 = \ln 4 + (2x-3)\ln 3$	A1	Or equivalent with no powers
	Solve for x	DM1	E.g. $x = \frac{\ln \frac{2}{81}}{\ln \frac{2}{9}}$ Allow with logarithms evaluated.
	Obtain 2.46 only	A1	
	Alternative Method for Question 7		
	Use a correct law of indices for a product or a power	*M1	E.g. $2^{x+1} = 2 \times 2^x$ Not available after incorrect manipulation, e.g. $6^{x+1} = 12^{2x-3}$. $4 = 2^2$ on its own is not enough.
	Obtain $\left(\frac{2}{9}\right)^x = \frac{2}{81}$	A1	OE with powers of x only, e.g. $6 \times 2^x = \frac{4}{27} \times 9^x$ Note: $(3^x)^2$ is not far enough.
	Solve for x	DM1	E.g. $x = \frac{\ln \frac{2}{81}}{\ln \frac{2}{9}}$ or $x = \ln_{\frac{2}{9}} \frac{2}{81}$. Allow with logarithms evaluated.
	Obtain 2.46 only	A1	
		4	

Question	Answer	Marks	Guidance
8	State or imply non-modular inequality $(3x+2)^2 < 3^2(2x-1)^2$, or pair of linear equations $(3x+2) = \pm 3(2x-1)$	B1	Allow '=', or any inequality sign.
	Make reasonable attempt at solving a 3-term quadratic, or solve two linear equations for x	M1	E.g. $27x^2 - 48x + 5 = 0$, $(9x-1)(3x-5) = 0$, $x = \dots$ Allow even if quadratic not given in a 3-term form. See guidelines document for solving a quadratic.
	Obtain critical values $x = \frac{1}{9}$ and $x = \frac{5}{3}$	A1	Allow '=', or any inequality sign.
	State final answer $x < \frac{1}{9}$, $x > \frac{5}{3}$ only	A1	Allow 'OR' but not 'AND'. Allow ' $\cup$ ' but not ' $\cap$ '. No marks can be scored if no working is seen.
	Alternative Method for Question 8		
	Obtain critical value $x = \frac{5}{3}$ from a graphical method, or by inspection, or by solving a linear equation or an inequality	B1	Allow '=', or any inequality sign.
	Obtain critical value $x = \frac{1}{9}$ similarly	B2	Allow '=', or any inequality sign.
	State final answer $x < \frac{1}{9}$, $x > \frac{5}{3}$ only	B1	Allow 'OR' but not 'AND'. Allow ' $\cup$ ' but not ' $\cap$ '. No marks can be scored if no working is seen.
		4	

Question	Answer	Marks	Guidance
9	Commence division and reach partial quotient of the form $3x^3 \pm 3x^2$ or $3x^4 - 2x^2 \equiv (x+1)(Ax^3 + Bx^2 + Cx + D) + Ex + F$, and reach $A = 3$ and $B = \pm 3$	M1	May be seen in synthetic division.
	Obtain quotient $3x^3 - 3x^2 + x - 1$ Do not ISW	A1	Don't need to state which is the quotient and which is remainder. However, if clearly muddled, then M1A1A0 for both expressions correct.
	Obtain remainder of 1	A1	Do not ISW. Allow e.g. $3x^3 - 3x^2 + x - 1 + \frac{1}{x+1}$ but NOT remainder = $\frac{1}{x+1}$.
	Alternative Method for Question 9		
	$f(-1) = 3 - 2 = 1 = \text{remainder}$	B1	Do not ISW.
	Use division or inspection or compare coefficients	M1	$3x^4 - 2x^2 - 1 \equiv (x+1)(3x^3 - 3x^2 + x - 1)$
	Obtain quotient $3x^3 - 3x^2 + x - 1$	A1	Do not ISW. Allow e.g. $3x^3 - 3x^2 + x - 1 + \frac{1}{x+1}$ but NOT remainder = $\frac{1}{x+1}$.
		3	

Question	Answer	Marks	Guidance
10	Use log quotient law, e.g. $\ln \frac{3}{5^x} = \ln 3 - \ln 5^x$ or log product law, e.g. $\ln 5^x + \ln(2^{3x-4})$	M1	May work in log to any base for first 3 marks.
	Use log power law, e.g. $\ln 5^x = x \ln 5$ or $\ln(2^{3x-4}) = (3x-4) \ln 2$	M1	Condone missing brackets if recovered at some point.
	Obtain correct expression for x in any exact form, e.g. $\frac{\ln 3 + 4 \ln 2}{3 \ln 2 + \ln 5}$	A1	
	Obtain final answer $\frac{\ln 48}{\ln 40}$	A1	Do not ISW. Final answer of $\log_{40} 48$ scores 3 marks only. No working award 0 marks.
	Alternative Method for Question 10		
	Rearrange to obtain $(2^3 \times 5)^x = 3 \times 2^4$	B1	OE
	Use log power law on an equation of the form $a^{bx} = c$ e.g. $x \ln(2^3 \times 5) = \ln(3 \times 2^4)$	M1	May work in log to any base for first 3 marks.
	Obtain correct expression for x in any exact form, e.g. $x = \frac{\ln(3 \times 2^4)}{\ln(2^3 \times 5)}$	A1	
	Obtain final answer $\frac{\ln 48}{\ln 40}$	A1	Do not ISW. Final answer of $\log_{40} 48$ scores 3 marks only. No working award 0 marks.
		4	

Question	Answer	Marks	Guidance
11	Obtain a 3-term quadratic in e^x	*M1	
	Obtain e.g. $3e^{2x} - 12e^x - 2 = 0$ or 3-term equivalent	A1	E.g. $3m^2 - 12m - 2 = 0$. '= 0' could be implied by subsequent working.
	Solve a 3-term quadratic to obtain a value for x or e^x	DM1	Need to get as far as a value for x or e^x .
	Obtain root $\frac{6 + \sqrt{42}}{3}$ or 4.16...	A1	OE Ignore second root if seen.
	Obtain answer 1.426 only	A1	CAO, must be 3 d.p. 0/5 for answer with no working. Second root must be rejected if seen.
		5	

Question	Answer	Marks	Guidance
12	Use correct trigonometric formulae to form an equation in $\tan x$ only, or an equation in terms of $\sin x$ and $\cos x$ only e.g. $\frac{3}{\tan x} - 4\left(\frac{1 - \tan^2 x}{2 \tan x}\right) = 3$ $\frac{3}{\tan x} - \frac{4}{\frac{2 \tan x}{1 - \tan^2 x}} = 3$ $3 \frac{\cos x}{\sin x} - 4 \frac{\cos^2 x - \sin^2 x}{2 \sin x \cos x} = 3$ $3 \cot x - 4\left(\frac{\cot^2 x - 1}{2 \cot x}\right) = 3$	*M1	Condone one slip in manipulating the original equation provided correct trig formulae used.
	Obtain a correct horizontal equation, in $\tan x$ or in $\cos x$ and $\sin x$, in any form	A1	E.g. $3 - (2 - 2 \tan^2 x) = 3 \tan x$ or $2 \sin^2 x - 3 \sin x \cos x + \cos^2 x = 0$.
	Reduce equation to a 3-term quadratic Allow if they square both sides and obtain a quartic in $\sin x$ or $\cos x$	A1	E.g. $2 \tan^2 x - 3 \tan x + 1 = 0$, $2 \sin^2 x - 3 \sin x \cos x + \cos^2 x = 0$. Allow if they leave the equation as a cubic with a factor of $\tan x$ and go on to give the correct solutions to the quadratic.
	Solve a 3-term quadratic to obtain a value for x The quadratic could be arrived at from a correct cubic Not available if they have an incorrect cubic with an incorrect quadratic factor.	DM1	If the initial equation is correct, then M1 is implied by a correct solution.
	Obtain one answer, e.g. $(x =) 45^\circ$	A1	
	Obtain a second answer, e.g. AWRT $(x =) 26.6^\circ$ and no other in the given interval, e.g. $x = 0$	A1	Ignore answers outside the given interval.
		6	

Question	Answer	Marks	Guidance
13	State that $1 - e^{-2x} = e^{-3}$	B1	OE, with ln removed.
	Use correct method to solve an equation of the form $e^{\pm 2x} = a$, where $a > 0$, and a reasonable attempt at the B1, for $\pm 2x \ln e$ or $\pm x \ln e$	M1	E.g. $[e^{-2x} = 1 - e^{-3}]$ $-2x = \ln(1 - e^{-3}) \dots$ OE. Can be numerical $\ln(1 - 0.049787) = \ln 0.9502$. Evidence of method must be seen.
	Obtain answer 0.0255	A1	CAO Must be 4 decimal places. No working seen scores 0.
	Alternative Method for Question 13		
	State that $1 - e^{-2x} = e^{-3}$	B1	OE, without ln.
	Rearrange to obtain an expression for e^x and solve an equation of the form $e^{\pm x} = a$, where $a > 0$, and a reasonable attempt at the B1, for x	M1	E.g. $\left[e^x = \sqrt{\frac{1}{1 - e^{-3}}}, e^x = \sqrt{\frac{e^3}{e^3 - 1}} \right],$ $x = \ln \sqrt{\frac{1}{1 - e^{-3}}}$ Can be numerical. Evidence of method must be seen.
	Obtain answer 0.0255	A1	CAO Must be 4 decimal places. No working seen scores 0.
		3	

Question	Answer	Marks	Guidance
14(a)	Substitute $x = 3$ or -3 into $p(x)$ and equate to 0 or into $p'(x)$ and equate to 72	M1*	
	Obtain $162 + 9a + 3b + 9 = 0$	A1	OE
	Obtain $162 + 6a + b = 72$	A1	OE
	Solve simultaneous equations to obtain either a or b after using $p(\pm 3) = 0$ and $p'(\pm 3) = 72$	DM1	
	Obtain $a = -11$ and $b = -24$	A1	
		5	

Question	Answer	Marks	Guidance
14(b)	Equate $(x-3)(6x^2 + Ax + B)$ to $6x^3 - 11x^2 - 24x + 9$ and obtain equations to solve for A and B	M1	Using <i>their a</i> and <i>their b</i> . $A - 18 = a = -11$, $B - 3A = -b = -24$, $-3B = 9$. $A = 7$ and $B = -3$.
	or divide $6x^3 - 11x^2 - 24x + 9$ by $x - 3$ and reach $6x^2 \pm 7x$		Or reach $6x^2 \pm (\text{their } a + 18)x$.
	$(x-3)(6x^2 + 7x - 3)$	A1	SOI
	Obtain $(x-3)(2x+3)(3x-1)$	A1	
			Special Case: If only $(x-3)(x+\frac{3}{2})(x-\frac{1}{3})$ or $(x-3)(2x+3)(3x-1)$ seen, SC B1 only (but can gain two marks in (c)).
		3	
14(c)	Obtain one correct region $x < -\frac{3}{2}$ or $\frac{1}{3} < x < 3$	B1 FT	Must be final answer not in working. FT is on the last two brackets (not $(x-3)$).
	Obtain both regions $x < -\frac{3}{2}$, $\frac{1}{3} < x < 3$	B1 FT	Allow $x < -\frac{3}{2}$ and $\frac{1}{3} < x < 3$. SC B1 for $x \leq -\frac{3}{2}$, $\frac{1}{3} \leq x \leq 3$. FT is on the last two brackets (not $(x-3)$). If incorrect factor or factors in (b) but correct regions here, allow SC B1 only.
		2	

Question	Answer	Marks	Guidance
15(a)	Take logarithms of both sides to obtain $\ln A + \ln y = x \ln b$	M1	Or three-term equivalent with no indices.
	Compare with $Y = mX + c$, $pY + qX + r = 0$	A1	AG Condone use of $y = mx + c$ in comparison.
		2	
15(b)	Form equations in A and b and solve for A or b	M1	$\ln A + 0.86 = 3.4 \ln b$ and $\ln A + 2.56 = 5.7 \ln b$. Or $Ae^{0.86} = b^{3.4}$ and $Ae^{2.56} = b^{5.7}$.
	Obtain $b = 2.1$ or $A = 5.2$	A1	Allow for 2 sf or more.
	Obtain $b = 2.1$ and $A = 5.2$	A1	Accept $A = 5.3$.
		3	

Question	Answer	Marks	Guidance
16(a)	State or imply $2 = \log_4 4^2$	B1	
	Use log power, product or quotient law e.g. $2\log_4 (3x - 1) = \log_4 (3x - 1)^2$	M1	E.g. $\log_4 (3x - 1)^2 - \log_4 4^2 = \log_4 \frac{(3x - 1)^2}{4^2}$
	Obtain a correct equation in any form, free of logs	A1	E.g. $16(2x + 1) = (3x - 1)^2$ or $9x^2 - 38x - 15 = 0$
		3	
16(b)	Solve 3-term quadratic	M1	
	Obtain answer 4.59 only	A1	$\frac{19 + 4\sqrt{31}}{9}$ OE. Accept AWR 4.59.
		2	